MARIAM ABU-RAHMA

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Portfolio: mariammabu.github.io

EDUCATION

Homestead High School, Sunnyvale, CA | *August 2023-Present*

- **GPA:** 4.0 (unweighted) | 4.47 (weighted)
- **Relevant Coursework:** AP Calculus BC, AP Physics 1, AP Chemistry, AP Computer Science A, AP Computer Science Principles
- **Interests:** Electrical Engineering, Mechanical Engineering, Robotics

SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, HTML/CSS
- **Hardware & Electrical:** Arduino microcontrollers, IR communication systems, motor control systems, circuit prototyping
- **Design & Engineering tools:** KiCad PCB design, Fusion 360, 3D printing, Github
- **Languages**: English, Arabic

PERSONAL PROJECTS

Lead Developer | AIxKumite: Computer Vision for Combat Sports | *June 2025 - Present*

Developing LLM-based systems using YOLOv8 and MediaPipie through coded JSON prompts to analyze karate matches and extract performance data to improve overall player performance. Using computer vision and pose-detection algorithms to identify techniques and evaluate athlete performance.

Hardware Engineer | Arduino IR Control System & PCB Design | *January 2026 - Present*

Designed and built a handheld infrared device to control LED systems using microcontroller programming. Integrated hardware circuits with software signal processing to transmit commands. Currently designing custom PCB using KiCad emphasizing clear signals and noise reduction for consistent signal processing.

Electrical Engineer | NFC Attendance Tracking System | April 2026 - May 2026

Designed and built an NFC-based attendance system using an ESP32 microcontroller and PN532 NFC reader to automate student check-ins at my karate dojo. Developed a WiFi-connected system that sends attendance data to Google Sheets through Google Apps Script integration. Improved user experience by adding LED feedback indicators for scan detection, processing status, successful check-ins, and error handling.

Portfolio: Engineering & Robotics Prototypes | mariammabu.github.io

Designed and built several robotics and electronics prototypes including:

- Stair-climbing robot: Rack and pinion system to move the robot up the steps
- LED lighting systems: Simple coin battery LED structures for 3D printed parts
- Ultrasonic levitator: Built using Open Source projects
- Additional electronics and robotics prototypes

RESEARCH EXPERIENCE

Research Intern | Basys.ai (Healthcare AI) *June - August 2026*

Worked with a team of 10 interns researching artificial intelligence systems used in healthcare. Studied how LLMs analyze patient electronic health records (EHR) and examined risks such as hallucinated medical outputs focusing on data training with poor quality and LLM's inconsistency in outputs. Collaborated with a group of 5-10 interns through daily meetings and technical discussions.

Lead Researcher | Journal of Emerging Investigators (JEI) *January - April 2024*
Co-Designed and conducted a cross-sectional survey study on hygiene behaviors after COVID-19 among high school students in Punjab, Pakistan and Santa Clara County, California (N=174) using Google Forms and social media outreach. Analyzed responses using two-proportion Z-tests in Excel to compare trends across gender and regions. Found that 28–36% of students no longer wore masks and 40–56% had returned to their pre-pandemic hygiene habits, showing that many students had stopped following COVID guidelines over time. Co-authored a 15-page manuscript and published it through the JEI peer-review process.

Robotics Program | MIT BeaverWorks (BWSI) *October - December 2025*
Programmed an autonomous vehicle using computer vision and line-following algorithms. Used OpenCV and HSV color detection to identify track markers and guide the car. Developed algorithms to adjust steering and speed based on detected errors. Team project earned 2nd place out of 10 teams and Academic Achievement recognition.

LEADERSHIP & TEACHING

Muslim Student Association (MSA) Event Coordinator | Homestead HS *2025 - Present*
Assist in coordination of events and activities such as budgeting, hosting speakers, and holding kahoot activities that support community engagement and leadership.

Karate Instructor & Mentor | Sunnyvale Martial Arts *January - April 2025*
Coached 20 younger athletes and assisted with training sessions, helping students improve sparring techniques using strategies I have learned from competing while exhibiting confidence during a tournament.

Founder & Curriculum Developer, STEM Outreach Initiative | Cairo, Egypt *July 2025*
Spearheaded an independent engineering outreach program, delivering technical workshops in Arabic to students in under-resourced communities. Curriculum featured motor-powered vehicles and LED circuitry to introduce fundamental Electrical and Mechanical Engineering concepts. Managed all logistics and instruction for 90-minute sessions, adapting complex engineering principles for students ages 6–12 to bridge gaps in local STEM education.

HONORS & INTERNATIONAL RECOGNITION

World Karate Federation (WKF) | Global Rank: #2

- **4x USA National Team Athlete:** Represent Team USA in elite international circuits
- **Gold Medalist:** 2025 Karate 1 (K1) Series – Monterey
- **Silver Medalist:** 2024 Karate 1 (K1) Series – Cancun
- **Bronze Medalist:** 2024 Pan American Championships – Brazil

Participant | ISSCC Women's Networking Initiative *February 2026*
Connected with professors from Berkeley, MIT, WPI, BU +, college students, and industry leaders in electrical engineering, and attended presentations regarding Integrated Circuit (IC) design and Moore's Law scaling.

5x Green & White Award Recipient | Homestead High School *2024 - 2026*
Recognized annually for demonstrating excellence and well roundedness in class settings while balancing international athletic commitments through math, science, engineering, and sports classes.

University of California Nominee in ELC Program | Homestead High School | 2026
Selected by University of California and Homestead High School for Eligibility in the Local Context (ELC), recognizing top academic performance within the graduating class and eligibility consideration for the University of California system.